Original Article

The Cost-effectiveness of Predonation Screening for Transfusion Transmissible Infections Using Rapid Test Kits in a Hospital-based Blood Transfusion Centre

Adedoyin Owolabi Dosunmu, Akinsegun Abduljaleel Akinbami, Ayobami Kamal Ismail, Modupe Adebimpe Olaiya[1], Ebele Ifeyinwa Uche, Igbinoba Kingsley Aile

Department of Haematology and Blood Transfusion, Lagos State University Teaching Hospital, Ikeja, [1]Lagos State Blood Transfusion Committee, Lagos Island Maternity Hospital, Lagos, Nigeria

Abstract

Objective: Blood transfusion practice emphasises safety, efficacy and appropriate use. These require cost-effective programme management. This study focused on the cost of screening for transfusion transmissible infections (TTI). **Methods:** This was a 1 year (2016) analysis of screening in a hospital-based transfusion centre. The cost of screening all blood donors by ELISA was compared to the cost of serial screening starting from rapid kit, taking into account, the estimated cost of blood bags prevented from discard after ELISA screening (attributable cost). The cost of voluntary donor drive plus cost of ELISA screening was compared with the present cost of screening. **Results:** A total of 5591 donors were screened for HIV, hepatitis B and C using the rapid kit, 291 donors were deferred (5.2%). A total of 5300 units were further screened by ELISA. A total of 435 blood units (8.2%) were discarded due to TTI positivity. TTI positivity rate was 12.98%. Only 2.36% were voluntary donors and among these 9.1% were TTI positive. The attributable cost of serial screening was 55,653.5 USD while that of screening by ELISA only was 55,910 USD. The attributable cost of rapid screening for only hepatitis B and then ELISA was 53,313.9 USD taking into consideration that 187 blood units would be prevented from undue discard. **Conclusions:** This analysis demonstrated that with proper donor selection, rapid screening for hepatitis B virus only before ELISA screening is more cost-effective. This will also reduce the waiting time for donors and counselling if HIV positive.

Keywords: ELISA screening, rapid test screening, transfusion transmissible infections

Introduction

Provision of safe blood is essentially based on voluntary nonremunerated blood donor, proper donor selection, transfusion transmissible infection (TTI) screening, blood component preparation and storage, proper compatibility testing and appropriate use of blood.[1] These processes require cost-effective management without compromising quality. In the developing countries, motivation and commitment to voluntary donation is poor. This is because the infrastructure, awareness and donor drive management are poorly funded. Most of the donations are therefore by hospital replacement donors.

Donor selection requires counselling, administration of questionnaires including a consent form and clinical examination. A replacement donor would have been tutored by the beneficiary through knowledgeable middlemen (touts) on how to respond to avoid deferment. The World Health Organization (WHO) therefore recommends 100% voluntary blood donation.[1] Within this group of voluntary donors, it has been established that the incidence rate of HIV and hepatitis C virus (HCV) are two times higher in first-time donors compared to repeated donors while it is four times greater with hepatitis B.[2] It is therefore essential to have a donor retention programme.

The WHO recommends the following screening tests to be done on all blood for transfusion: HIV 1 and 2, hepatitis B

Access this article online

Quick Response Code:

Website: www.npmj.org

DOI: 10.4103/npmj.npmj_33_17

Address for correspondence: Dr. Adedoyin Owolabi Dosunmu, Department of Hematology and Blood Transfusion, Lagos State University Teaching Hospital, Ikeja, Lagos, Nigeria.
E-mail: doyin_dosunmu@yahoo.com

This is an open access article distributed under the terms of the Creative Commons Attribution-NonCommercial-ShareAlike 3.0 License, which allows others to remix, tweak, and build upon the work non-commercially, as long as the author is credited and the new creations are licensed under the identical terms.

For reprints contact: reprints@medknow.com

How to cite this article: Dosunmu AO, Akinbami AA, Ismail AK, Olaiya MA, Uche EI, Aile IK. The cost-effectiveness of predonation screening for transfusion transmissible infections using rapid test kits in a Hospital-based blood transfusion centre. Niger Postgrad Med J 2017;24:162-7.

surface antigen (HBS Ag), HCV antibody and syphilis.[1] Based on the geographical zone, other tests such as malaria parasite, Trypanosoma cruzi and human T lymphotropic virus 1 may be added while in immunocompromised patients, cytomegalovirus screening is recommended.[1] The minimal requirements are serological tests, but these were found to be inadequate due to the long window period before seroconversion, mutant forms of antigens or laboratory performance errors.[1,3] Some centres have therefore added surrogate tests such as serum alanine aminotransferase (ALT) and hepatitis B core antibodies. In advanced countries, nucleic acid amplification tests (NAT) have been introduced, and this has reduced the window period to about 11 days in mini-pool (16–24 samples) NAT assays and 5–6 days in individual donor NAT tests. These tests are expensive but have reduced the residual risk of HIV by 50%, HCV by 72% and HBV by 42% through reduction of the window period and eliminating the effect of variants viruses.[4,5]

Whatever be the practice, the cost-effectiveness requires continuous evaluation. ALT test has subsequently been dropped, anti–HB core antibody test is optional, the possibility of doing individual donor NAT test is being considered, and automated NAT test that can handle more tests of multiple viruses simultaneously may provide the economy of scale without compromising quality.[1,6] Irrespective of the screening method, there is need to have a quality system in place which should include National or State validation of equipment and kits, accreditation of screening agencies with evidence of efficient stock storage including transportation management and effective state oversight.[1] In 2004, Lagos State Government passed into law the establishment of the committee for blood transfusion to perform the functions stated above among other functions.

The Lagos State University Teaching Hospital (LASUTH) blood transfusion centre bleeds donors after a group counselling session followed by administration of questionnaire and consent forms. These allow for self-deferral. The prospective donors are subjected to clinical assessment which includes blood pressure measurement, weight, haemoglobin level, rapid screening for HIV, hepatitis B and C with a brief physical examination. Blood so collected are sent for screening by an agency that screens the blood for HIV antibody and p24 antigen, HBs antigen, HCV antibody and syphilis by ELISA. Positive blood bags are retained by this agency for discard. The hospital blood transfusion centre pays a fixed amount for this service.

The purpose of this study is to examine the cost-effectiveness of the initial rapid test screening and then advice the hospital for possible changes. It is the hope of the authors that other transfusion centres will benefit from this study.

Methods

This was a retrospective programme management study of the cost-effectiveness of screening for TTI in blood donors at the LASUTH during the year 2016. The donors had their consent for TTI test taken after counselling and they were reassured of confidentiality according to the provision of the State Transfusion committee. Data were collated from January 1 to December 31, 2016.

A volume of 4 mls of blood was collected from each donor at registration into an EDTA sample bottle. The haematocrit was done using the haematocrit centrifuge and reader after which the samples were spun and plasma released for rapid TTI screening. The rapid screenings were done using the Determine HIV 1/2 test kit by Alere Medical Co. Ltd. Japan for HIV 1/2 screening and DiaSpot reagents from the USA for hepatitis B and C. Positive samples were not confirmed as such donors were permanently deferred and referred with the consent of the donor to the appropriate clinic for confirmation and treatment.

The HIV ELISA screening was done by the sandwich technique to detect the p-24 antigen and antibodies to HIV 1, epitopes of the group O subtypes and HIV-2 envelope protein using the Genscreen™ Ultra HIV Ag-Ab kit by Bio-Rad, France. Hepatitis B screening was done using a double antibody sandwich ELISA method for hepatitis B antigen by Dialab Produktion, und Vertrieb von chemisch-technischen Produkten und Laborintrumenten Gesellschaft MBH with a sensitivity of 100% and a specificity higher than 99.5%.[1] Hepatitis C screening was done using ELISA to determine antibodies against antigens corresponding to the core and non-structural regions on hepatitis C by Dialab Produktion with a sensitivity of 99.79% and specificity of 99.55%.[1] The syphilis test was done using microplates coated with recombinant antigen for Treponema pallidum to detect total antibodies (IgA, IgG and IgM) in the serum or plasma by Dialab Produktion.[1]

The number of donors that presented for donation was captured, the number of donors positive to HIV, hepatitis B and C by rapid screening were noted. The number of blood bags sent for ELISA testing, and the numbers of positive blood units by ELISA tests were noted. The cost of the rapid kits, the cost of syringes and sample bottles for collection of initial sample for the rapid test were summed up to obtain the cost for rapid screening.

The demography of the donors was obtained from the questionnaire filled by them. The costing was based on the direct cost of the reagents and consumables directly linked to the tests only while utilities, man hours, equipment purchase and maintenance were considered as non-variable fixed costs.

The results were expressed in means or medians and percentages. Monetary calculations were done in Dollar (USD) at an average rate of 1 Dollar (USD) to 250 Naira (N). During the period of this study, the value of one Dollar varied between 195 and 300 Naira. Degrees of association were done using a contingency table and the Chi-square test. The statistical programme used was the GraphPad InStat 3.

Results

There were 5591 donors who presented for rapid screening, the number of females was 251 (4.5%) and the age range was 18 to

62 years with a median of 29 years. The number of voluntary donors was 132 (2.36%).

At initial pre-donation screening using rapid kits; 229 (4.1%) were hepatitis B positive while 7 (0.13%) and 55 (0.98%) were HIV and hepatitis C positive. A total of 291 donors (5.2% or 1 in 19 donors) were permanently deferred on account of TTI. If hepatitis B virus rapid test was the only test done, 1 in 24 donor would have been permanently deferred [Table 1]. The number of blood units sent for ELISA screening was 5300, and a total of 435 (8.2% or 1 in 12) units of blood were retained for discard. Assuming that the ELISA kit would identify all the donor samples positive by rapid kit, a total of 726 units of blood (12.98% or 1 in 7 blood units) would have been retained by the agency for discard if there were no rapid screening. Similarly, if only hepatitis B virus was the rapid test done, 5362 blood units would have had ELISA tests done and out of which 497 units (9.2% or 1 in 11 units) would have been discarded [Tables 2 and 3].

There were 132 voluntary donors whose blood units were screened with ELISA kits and 12 were positive (9.1%). Therefore, if, there were no rapid screening but proper voluntary donor selection was done, 3.88% (12.98% that were positive overall minus 9.1% positive among voluntary blood donors) of the blood units would have been safe for transfusion because TTI positivity is less by 3.88% among voluntary donor compared to relative donors. The number of HIV/hepatitis/syphilis co-infections was 22 (0.42%) with a statistically significant association ($P < 0.0001$) [Table 4].

The amount paid to the screening centre was 10 USD per unit of blood screened. Therefore, the total cost of screening of all the donors without rapid screening would have been 55,910 USD (10 USD × 5591 donors = 1A in Table 5). The cost of rapid screening of all the donors was 6,261.9 USD at a rate of 1.12 USD per donor (estimated cost of kits and consumables × 5591 donors). If this is added to the cost of ELISA screening after rapid screening which was 53,000 USD (10 USD × 5300 donors), the total cost of screening that includes rapid screening should be 59,261.9 USD (cost of rapid screening + cost of ELISA screening). A total of 291 units of blood that were not collected from deferred donors would have been discarded after ELISA screen at the cost of 12.4 USD per bag (cost of blood bag and ELISA screening). The cost of the extra bags that would have been discarded would be 3608.4 USD (291 × 12.4 USD). The total cost of serial screening minus the saving (attributable cost) is therefore 55,653.5 USD (total cost of serial screening minus cost saved by rapid test). The blood bank, therefore, saved 356.48 USD (cost of ELISA screening of all donors minus attributable cost of serial screening) by doing serial screening using rapid test before ELISA.

The rapid hepatitis B test picked about 79% of all positive donors screened by rapid tests. Assuming only the hepatitis B screen test was done at about 0.36 USD per tests, the cost of rapid screening would have been 2,012.76 USD (5591 × 0.36 USD).

Table 1: Rapid tests results ($n=5591$)

Test	Number positive	Percentage	Percentage of total positive
HIV	7	0.13	2.4
HBV	229	4.1	78.7
HCV	55	0.98	18.9
Total	291	5.2	100

HBV: Hepatitis B virus, HCV: Hepatitis C virus

Table 2: Elisa screen result ($n=5300$)

Test	Number positive	Percentage	Percentage of total positive
HIV	133	2.49	30.6
HBV	187	3.53	43
HCV	64	1.2	14.7
Syphilis	51	0.96	11.7
Total	435	8.2	100

HBV: Hepatitis B virus, HCV: Hepatitis C virus

Table 3: Total result ($n=5591$)

Tests	Number positive	Percentage	Percentage by rapid test
HIV	140	2.5	0.05
HBV	416	7.44	55
HCV	119	2.13	46
Syphilis	51	0.91	NA
Total	726	12.98	40

HBV: Hepatitis B virus, HCV: Hepatitis C virus, NA: Not available

Table 4: Probability of co-infections: $\chi^2=27.105$ ($P=0.0001$)

Test	Co-infections	No co-infections	Total	Percentage
HIV	6	127	133	4.5
HBV	18	169	187	9.6
HCV	4	60	64	7.8
Syphilis	15	36	51	29.4

HBV: Hepatitis B virus, HCV: Hepatitis C virus

Using similar principles to calculate the attributable cost of HBV rapid screen only (53,313.9 USD), the blood bank would have saved 1876.4 USD.

Since 3.88% of extra blood units would have been provided safe for transfusion if all the units were donated by voluntary blood donors, extra 217 units of blood would have been sold at 18 USD per unit of blood with a total cost of 3,906 USD (18 × 217 blood units). At present, voluntary blood donation drive costs about 7.2 USD per donor. The total cost of donor drive for all donors if they were voluntary blood donors would, therefore, be 40,255.2 USD. Voluntary blood donation would therefore cost 36,605.7 USD extra. To be cost-effective, the programme needs to spend 40,255.2 − 36,605.7 USD (3,649.5 USD) on 5591 donors. Therefore, a sum of 0.65 USD

Table 5: Costing

Cost items	*n*	Unit cost	Total cost	Attributable cost (in USD)
1. Cost of ELISA only	5591	10	55,910	55,910 (1 A)
2. Cost of rapid tests	5591	1.12	6261.92	
3. Cost of ELISA after rapid test	5300	10	53,000	
4. Total cost of rapid + ELISA test	5591		59,261.90	
5. Cost of blood bag saved	291	12.4	3608.4	55,653.5 (line 4-5=5 A)
6. Amount saved by rapid screen			256.48	
7. Cost of HBV test only	5591	0.36	2012.76	
8. Cost of ELISA after HBV test	5362	10	53,620	
9. Total cost of HBV + ELISA tests	5591		55,632.76	
10. Cost of blood bags saved after HBV test	187	12.4	2318.8	53,313.9 (line 9-10=10 A)
11. Amount saved with HBV + ELISA tests			1876.40	
12. Cost of donor drive	5591	7.2	40,255.2	
13. Cost of donor drive + ELISA test	5591		96,165.2 (line 12+1)	
14. Cost of blood saved by voluntary donation=3.88%	217	18	3906	92,259.2 (line 13-14=14 A)
15. Excess cost on donor drive			36,605.7 (5-14 A)	
16. Donor drive efficient cost	5591		3649.5 (12-15)	
17. Cost donor drive	1		0.65 (16/5591)	

HBV: Hepatitis B virus

per donors (or 5% of cost blood bag plus screening) is expected be cost-effective in a donor drive [Table 5].

Discussions

Most of the donors were young male who were donating for friends or relations (median age was 29 years and females were 4.5%). The ideal blood donor is a regular non-remunerated voluntary blood donor.[1] In practice, voluntary blood donors form a very small percentage (2.3%). Most of them were probably not regular blood donors since the prevalence (9.1%) of TTI was very high.[2] Although this may also be a reflection of the high prevalence of HIV and hepatitis virus in the society.[7,8] A voluntary blood donor drive costs about 7.2 USD per donor. It requires appealing to groups, visiting them, arranging for donor date, mobilising a team to execute the function and provision of gift items. The blood bank would have spent over 40,000 USD to save 4000 USD [Table 5, rows 12 and 14]. The present donor drive is therefore not only inadequate, but it is also not cost-effective. Donor drive is cost-effective if cost can be limited to about 0.65 USD per donor (estimated cost-effective value/number of all donors). This might be achieved if more voluntary donors were recruited to have an economy of scale. The high prevalence of TTI among voluntary blood donors needs to be reduced probably by focusing on individuals rather than groups because donor selection in a group seems to have similar challenges as replacement donors. It is therefore desirable to have an improved donor drive that is individualised, that ensures donor retention and that recruits more so as to be cost-effective.[2-4,9,10]

During the period of study, the prevalence of TTIs among donors in LASUTH after donor selection were 2.5% for HIV, 7.44% for hepatitis B, 2.13% for hepatitis C and 0.91% for syphilis [Table 3]. These are low compared to that in the general population which are 3.8%–14% for HIV, above 8% for hepatitis B, 4%–11% for hepatitis C and 1.6% for syphilis.[11-15] Studies among blood donors in a similar geopolitical region demonstrated the prevalence of 18.6%, 3.1%, 6% and 1.1% for hepatitis B, HIV, hepatitis C and syphilis, respectively in 2009.[16]

The prevalence of TTI obtained in this study, is however, slightly higher than the prevalence in developing countries in Asia and similar to data from urban areas in Nigeria which justifies the use of donor education, questionnaires and clinical examination to encourage self-deferrals with improvement in donor selection over the years.[17-19]

The p24/antibody ELISA kit identified 133 samples earlier screened negative by rapid kit as HIV positive [Tables 2 and 3]. There is however, the possibility of high false positive rate due to cross-reactivity of the p24 antigen and antigen on herpes simplex virus type 1 strain 17.[20] In general, 726 (about 13%) donors were positive and rapid screening detected 40% of these thus saving 291 units that would have been bled and screened by ELISA at a rate of 12.4 USD per unit (i.e., cost of blood bag plus TTI screening). The transfusion centre by so doing saved 256.48 USD compared to 1,876.4 USD that would have been saved if only HBS Ag rapid screening was done [Table 5]. A similar study by Shittu *et al*. 2014 at Ilorin implied that predonation screening for HIV and hepatitis B and C was not cost-effective.[21]

Over 75% of permanent deferment after rapid screening was due to hepatitis B virus. This may be due to the high frequency in the community and shorter period of seroconversion as a result of the antigen testing. About 57% of blood units for discard were due to hepatitis B virus despite the rapid antigen testing. This highlights the diversity in the surface antigen. Some mutant surface antigen epitopes ('a' determinants) were

probably missed by rapid screening while some more were detected by ELISA screening but the ultimate is nucleic acid amplification testing.[22,23] Hepatitis B infection in childhood tends to progress to chronicity than in the adult.[24] The inclusion of hepatitis B vaccine as part of expanded programme on immunisation and prevention of vertical transmission with passive immunisation of positive pregnant women should reduce this high endemic nature in Nigeria over the years.[23]

There were 22 co-infections (0.42%) and HBV had the highest co-infection rate (6.4%, 2.25% and 1.6% with syphilis, HIV and HCV, respectively). For every 10 sample positive to hepatitis B virus, there is the possibility of avoiding one other TTI. The frequency of co-infection is much lower than the study by Ojo *et al*. in pregnant women where the co-infection rate with HIV was 22.2%.[25] The co-infection rate among HIV patients in North Central Nigeria was also higher at 20.6%.[26] However, the co-infection rate in this study is similar to the rate of 0.77% seen in blood donors in Cameroon.[27] In addition, in this study, there was a strong association of co-infections ($P < 0.0001$). This is understandable because all the TTIs are transmissible by similar means and it is similar to the report from Ethiopia by Belay Tessema *et al*. where there was association between HIV, HCV and syphilis.[28] The high prevalence of hepatitis B, the higher amount saved and the higher co-infection rate qualifies it to be used as the only rapid kit screening marker for donors. Moreover, the presence of mutant antigen forms requires a serial screening test for hepatitis B.

Limitations of study

The costing of HBV only screening and that of 100% voluntary donation were derived from established study data since such practices were not in place. The study made some assumptions that the cost of reagent kits and consumables were constant but there were differences in the costs during the year and there were periods the HIV kits were given to the blood bank free by the State or 'PEPFAR' implementing partners; that the ELISA screening would have detected all the positive samples that were detected by rapid screen and that the false positive rate with ELISA screening is not significantly high. These assumptions were necessary in order to compute the cost of alternative practices.

Conclusions

It is economical to do rapid predonation screening where the prevalence of an infection is high and the possibility of subtypes and mutants are high. In our case, HBS Ag screening is recommended. This should be combined with cost-efficient voluntary donor drive, donor retention strategies and good donor selection to make blood for transfusion safe and cost-effective.

Acknowledgement

We appreciate the dedication and cooperation of the staff of the donor clinic and the TTI screening laboratory. We also thank Mr. Ebenezer Gboyemi for data collation.

Financial support and sponsorship

Nil.

Conflicts of interest

There are no conflicts of interest.

References

1. Screening Donated Blood for Transfusion-Transmissible Infection: Recommendations. WHO; 2010. Available from: http://www.who.int/bloodsafety/Screening.[Last accessed on 2017 Mar 20].
2. Stramer SL, Glynn SA, Kleinman SH, Strong DM, Caglioti S, Wright DJ, *et al.* Detection of HIV-1 and HCV infections among antibody-negative blood donors by nucleic acid-amplification testing. N Engl J Med 2004;351:760-8.
3. Busch MP, Kleinman SH, Jackson B, Stramer SL, Hewlett I, Preston S. Committee report. Nucleic acid amplification testing of blood donors for transfusion-transmitted infectious diseases: Report of the Interorganizational Task Force on Nucleic Acid Amplification Testing of Blood Donors. Transfusion 2000;40:143-59.
4. Zou S, Dorsey KA, Notari EP, Foster GA, Krysztof DE, Musavi F, *et al.* Prevalence, incidence, and residual risk of human immunodeficiency virus and hepatitis C virus infections among United States blood donors since the introduction of nucleic acid testing. Transfusion 2010;50:1495-504.
5. Kleinman SH, Kuhns MC, Todd DS, Glynn SA, McNamara A, DiMarco A, *et al.* Frequency of HBV DNA detection in US blood donors testing positive for the presence of anti-HBc: Implications for transfusion transmission and donor screening. Transfusion 2003;43:696-704.
6. Buseh MP, Korelitz JJ, Klei SH, Lee SR, AuBuchon JP, Schreiber GB. Declining value of alanine transferase in screening of blood donors to prevent post transfusion hepatitis B and C infctions. The retrovirus Epidemiology Donor study. Transfusion 1995;35:903-10.
7. Shepard CW, Simard EP, Finelli L, Fiore AE, Bell BP. Hepatitis B virus infection: Epidemiology and vaccination. Epidemiol Rev 2006;28:112-25.
8. Uneke CJ, Ogbu O, Inyama PU, Anyanwu GI, Njoku MO, Idoko JH. Prevalence of hepatitis-B surface antigen among blood donors and human immunodeficiency virus-infected patients in Jos, Nigeria. Mem Inst Oswaldo Cruz 2005;100:13-6.
9. Durro V, Koraqi A, Saliasi S. Trends in the prevalence of transfusion-transmissible infections among blood donors in Albania. Clin Lab 2010;56:591-5.
10. Atar Chi Z, Ghatouri M, Hajibaygi B, Asaari SH, Alavian SM. Donor deferral and blood borne infection in blood donor in Tehran. Sci J Blood Transfus Organ 2006;2:323-64.
11. Chikwem JO, Mohammed I, Okara GC, Ukwandu NC, Ola TO. Prevalence of transmissible blood infections among blood donors at the University of Maiducuri Teaching Hospital, Maiduguri, Nigeria. East Afr Med J 1997;74:213-6.
12. Ejiofor OS, Emechebe GO, Igwe WL, Ifeadike CO, Ubajaka CF. Hepatitis C virus infection in Nigeria. Niger Med J 2010;51:173-6.
13. Olokoba AB, Salawu FK, Danburam A, Olokoba LB, Midala JK, Badung LH, *et al*. Hepatitis B virus infections among pregnant women in North Eastern Nigeria – A call for action. Niger J Clin Pract 2011;14:10-3.
14. Egah DZ, Banwal EB, Audu ES, Iya D, Mandong BM, Anele AA, *et al*. Hepatitis B s antigen, heatitis C virus and human immunodeficiency virus antibodies in a low risk donor group, Nigeria. East Mediterr Health J 2007;13:954-61.
15. Ani MN, Agwale SM. Human immunodeficiency virus infection in Nigeria. Braz J Infect Dis 1998;2:143-59.
16. Buseri FI, Muhibi MA, Jeremiah ZA. Sero-epidemiology of transfusion-transmissible infectious diseases among blood donors in Osogbo, South-West Nigeria. Blood Transfus 2009;7:293-9.
17. Arora D, Arora B, Khetarpal A. Seroprevalence of HIV, HBV, HCV and syphilis in blood donors in Southern Haryana. Indian J Pathol Microbiol 2010;53:308-9.

18. Koate BB, Buseri FI, Jeremiah ZA. Prevalence of hepatitis C virus among blood donors in River State, Nigeria. Transfus Med 2005;15:449-51.
19. Zou S, Fujii K, Johnson S, Spencer B, Washington N, Iv EN, *et al.* Prevalence of selected viral infections among blood donors deferred for potential risk to blood safety. Transfusion 2006;46:1997-2003.
20. Langedijk JP, Vos WF, van Doornum GJ, Rotman HA, Meloen RH, Huisman JG. Identification of cross-reactive epitopes recognized by HIV-1 false-positive sera. AIDS 1992;6:1547-8.
21. Shittu AO, Olawumi HO, Adewuyi JO. Predonation screening of blood for transfusion transmissible infections: The gain and pains experienced at a resource limited blood bank. Ghana Med J 2014;48:158-62.
22. Levicnik-Stezinar S. Hepatitis B surface antigen escape mutant in a first time blood donor potentially missed by a routine screening assay. Clin Lab 2004;50:49-51.
23. Emechebe GO, Emodi II, Ikefuna AN, Ilechukwu GC, Igwe WC, Ilechukwu CA. Hepatitis B virus infection in Nigeria. A review. Niger Med J 2009;50:18-22.
24. McMahon BJ, Alward WL, Hall DB, Heyward WL, Bender TR, Francis DP, *et al.* Acute hepatitis B virus infection: Relation of age to the clinical expression of disease and subsequent development of the carrier state. J Infect Dis 1985;151:599-603.
25. Ojo OT, Jagun OE, Ikhile MU, Olatunji PO, Bakare BY. Seroprevalence and co-infection of HIV, HBVand syphilis among booked pregnant women at Olabisi Onabanjo University Teaching Hospital. Niger Med J 2016;70:34-8.
26. Forbi JC, Gabadi S, Alaba R, Speredoln HO, Pam CR, Entonu PE, *et al*. The role of triple infections with hepatitis B virus, hepatitis C virus and human immunodeficiency virus type 1 on CD4 lymphocytes level in the highly HIV infected population of North Central Nigeria. Mem Inst Oswaldo Cruz 2007;102:535-7.
27. Fouelifack Ymele F, Keugoung B, Fouedjio JH, Kouam N, Mendibi S, Dongtsa Mabou J. High rates of hepatitis B and C and HIV infections among blood donors in Cameroon: A proposed blood screening algorithm for blood donors in resource-limited settings. J Blood Transfus 2012;2012:458372.
28. Tessema B, Yismaw G, Kassu A, Amsalu A, Mulu A, Emmrich F, *et al*. Seroprevalence of human immunodeficiency virus, hepatitis B virus, hepatitis C virus and syphilis infections among blood donors at Gondar University Teaching Hospital, Northwest Ethiopia: Declining trend over a period of five years. BMC Infect Dis 2010;10:111.